PHB 243b: Practicum in Biostatistics

Lecture 13: Introduction to ADNI and Modeling MCI Progression

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Lecture 13: ADNI and MCI Progression

Today's Topics

1. The Alzheimer's Disease Neuroimaging Initiative (ADNI)
2. Understanding Mild Cognitive Impairment (MCI)
3. MCI to AD Conversion: The Clinical Problem
4. Project 2 Hypothesis: The Grassi et al. Approach
5. Key Predictors and Features
6. Machine Learning Methods Overview
7. Project 2 Requirements

Part I: The ADNI Study

What is ADNI?

Alzheimer's Disease Neuroimaging Initiative (ADNI)

A longitudinal multicenter study designed to develop clinical, imaging, genetic, and biochemical biomarkers for early detection and tracking of Alzheimer's disease.

- **Started**: 2004 (ADNI-1)
- **Phases**: ADNI-1, ADNI-GO, ADNI-2, ADNI-3
- **Sites**: 57 sites across US and Canada
- **Data**: Publicly available at adni.loni.usc.edu

ADNI Study Design

Phase	Years	N	Focus
ADNI-1	2004-2009	819	Establish biomarker methods
ADNI-GO	2009-2011	131	Early MCI (EMCI) cohort
ADNI-2	2011-2016	790	Expanded biomarkers
ADNI-3	2016-2022	696	Tau PET, digital biomarkers
ADNI-4	2022-present	596	Diversity, novel biomarkers

Total enrollment: N = 3,032 participants (Donohue et al., 2026)

Diagnostic Groups in ADNI

Group	Abbreviation	Definition
Cognitively Normal	CN	No memory complaints, normal cognition
Early MCI	EMCI	Mild memory concern, CDR = 0.5
Late MCI	LMCI	More pronounced deficits, CDR = 0.5
Alzheimer's Disease	AD	Dementia diagnosis, CDR 0.5

The distinction between EMCI and LMCI is based on education-adjusted delayed recall scores on the Logical Memory II subscale.

The alzverse: R Packages for ADNI Data

New in 2026: The `alzverse` ecosystem provides curated, analysis-ready ADNI data with reproducible vignettes (Donohue et al., 2026).

Package	Description
`ADNIMERGE2`	Updated ADNI data with pharmaverse/ADaM workflows
`A4LEARN`	A4 trial and LEARN observational study
`alzverse`	Meta-package integrating multiple studies

- Cited in ~250 publications since 2013
- Includes built-in documentation and analysis vignettes
- Supports reproducibility via `renv` and Docker

Installing ADNIMERGE2

```r
# Download from ADNI website (requires Data Use Agreement)
# https://adni.loni.usc.edu

# Install the downloaded package
install.packages("ADNIMERGE2_x.x.x.tar.gz", repos = NULL)
library(ADNIMERGE2)

# Access subject-level data (ADaM format)
data(ADSL)  # Subject-level analysis dataset
data(ADQS)  # Questionnaire/cognitive scores
```

Key variables in ADNIMERGE2:

- Demographics: AGE, SEX, RACE, ETHNIC, EDUCAT
- Diagnosis: DX, DX_BL (baseline diagnosis)
- Cognition: MMSE, ADAS11, ADAS13, RAVLT scores, LDELTOTAL
- Function: FAQ, CDR-SB
- Imaging: Hippocampus, Ventricles, WholeBrain
- Genetics: APOE genotype, amyloid status

ADNI Baseline Characteristics (N = 3,032)

Characteristic	Value
Age, mean (SD)	72.2 (7.6) years
Sex	50% Female, 50% Male
Education, mean (SD)	16.0 (2.8) years
Race: White	81%
Race: Black/African-American	13%
Ethnicity: Hispanic/Latino	6.4%

Baseline Diagnosis:

- Cognitively Normal (CN): 40% (n = 1,215)

- Mild Cognitive Impairment (MCI): 44% (n = 1,338)
- Dementia (DEM): 16% (n = 477)

ADNI Data Structure

```r
library(ADNIMERGE2)
data(ADSL)  # Subject-level (one row per subject)
data(ADQS)  # Cognitive scores (one row per visit per test)

# Examine structure
dim(ADSL)
names(ADSL)[1:15]

# Baseline diagnoses
table(ADSL$DX_BL)
```

Key insight: ADSL has one row per subject; ADQS has multiple rows per subject for longitudinal cognitive assessments.

Part II: Understanding MCI

What is Mild Cognitive Impairment?

MCI Definition (Petersen Criteria)

- Subjective memory complaint (corroborated by informant)
- Objective memory impairment for age/education
- Preserved general cognitive function
- Intact activities of daily living
- Not demented

MCI represents a transitional state between normal aging and dementia.

MCI Subtypes

Subtype	Description	AD Risk
Amnestic MCI (aMCI)	Memory impairment primary	Higher
Non-amnestic MCI	Other cognitive domains	Variable

Subtype	Description	AD Risk
Single-domain	One cognitive domain affected	Lower
Multi-domain	Multiple domains affected	Higher

ADNI focuses primarily on **amnestic MCI** (single and multi-domain).

The Clinical Problem

Not all MCI patients progress to AD

- ~10-15% of MCI patients convert to AD annually
- Some remain stable (stable MCI, sMCI)
- Some revert to normal cognition

Clinical need: Identify which MCI patients will progress to AD

- Treatment planning
- Clinical trial enrollment
- Patient and family counseling
- Healthcare resource allocation

MCI to AD Conversion Rates

From ADNI data (Grassi et al., 2019):

Outcome	N	Percentage
Stable MCI (sMCI)	353	64.2%
Converter to AD (cAD)	197	35.8%
Total	550	100%

Over a 3-year follow-up period, approximately **36% of MCI patients** in ADNI converted to AD dementia.

Part III: Project 2 Hypothesis

The Grassi et al. (2019) Study

Your Project 2 Hypothesis

Replicate and extend the approach of Grassi et al. (2019) to predict conversion from MCI to AD using machine learning methods applied to baseline clinical and cognitive data.

Reference: Grassi M, et al. (2019). A Clinically-Translatable Machine Learning Algorithm for the Prediction of Alzheimer's Disease Conversion in Individuals with Mild and Premild Cognitive Impairment. *J Alzheimers Dis*, 67(4):1353-1365.

Study Design: Grassi et al.

Cohort Selection:

- 550 MCI subjects from ADNIMERGE
- Baseline data only (no longitudinal features)
- 3-year follow-up for conversion assessment
- Site-independent train/test split

Outcome:

- Binary: Converter (cAD) vs. Stable (sMCI)
- Conversion = diagnosis of AD at any follow-up visit

Key Results: Grassi et al.

Metric	Value
AUROC	0.88
Sensitivity	77.7%
Specificity	79.9%
PPV	71.9%
NPV	84.3%

Interpretation: The model correctly identifies ~78% of future converters while maintaining ~80% specificity for stable patients.

Grassi et al. Table 2: Cohort Characteristics

Grassi et al. Prediction of Conversion to Alzheimer's

TABLE 2 | Descriptive statistics.

Continuous predictors		Non-converters		Converters		Missing values	
		Mean	**S.D.**	**Mean**	**S.D.**	***N***	**%**
Age		72.42	7.54	74.19	6.88	/	/
Years of education		16.18	2.74	15.74	2.83	/	/
CDRSB		1.26	0.70	1.95	1.01	/	/
ADAS11		8.67	3.78	12.94	4.26	1	0.18
ADAS13		13.89	5.81	21.05	5.72	3	0.55
ADASQ4		4.61	2.35	7.16	2.04	/	/
MMSE		28.01	1.71	26.85	1.72	/	/
RAVLT-I		37.84	10.47	28.05	6.74	/	/
RAVLT-L		4.76	2.59	2.90	2.11	/	/
RAVLT-F		4.37	2.46	5.20	2.30	/	/
RAVLT-PF		51.09	30.92	78.20	28.04	/	/
LDT		6.84	3.12	3.59	2.89	/	/
DIGIT		40.24	10.42	34.86	11.02	290	52.73
TMTBT		100.30	49.56	141.24	79.66	4	0.73
FAQ		1.76	2.75	5.81	5.00	4	0.73
Categorical predictors		**Non-converters**		**Converters**		**Missing values**	
		N	**%**	***N***	**%**	***N***	**%**
Sex	Male	220	62.32	118	59.90	/	/
	Female	133	37.68	79	40.10		
Subtype of MCI	Early	196	47.88	22	11.17	/	/
	Late	184	52.12	175	88.83		
Marital status	Never married	6	1.70	3	1.52	3	0.55
	Married	267	75.64	161	81.73		
	Divorced	35	9.92	13	6.60		
	Widowed	42	11.90	20	10.15		

S.D, Standard Deviation; N, numbers of subjects.

Figure 1: Grassi et al. Table 2

Your goal: Reproduce a similar table for your MCI cohort, comparing converters (cAD) vs. stable (sMCI) subjects.

Why This Approach?

Clinical Translatability:

1. Uses only baseline data (no follow-up required)
2. Relies on standard clinical assessments
3. No advanced imaging required (MRI, PET)
4. Applicable in community clinical settings

The goal is a **clinically practical** prediction tool, not maximum possible accuracy with expensive biomarkers.

Part IV: Key Predictors

Feature Categories

Grassi et al. used four categories of baseline features:

Category	Examples
Sociodemographic	Age, education, sex, marital status
Clinical Scales	CDR-SB, FAQ
Neuropsychological	MMSE, ADAS, RAVLT, LDELTOTAL, TMTB
MCI Subtype	EMCI vs. LMCI

Note: Imaging and genetic features were intentionally excluded to maximize clinical applicability.

Sociodemographic Features

```r
# Extract sociodemographic features
demo_vars <- adnimerge |>
  filter(VISCODE == "bl") |>
  select(RID, AGE, PTEDUCAT, PTGENDER, PTMARRY)

head(demo_vars)
```

Variable	Description	Coding
AGE	Age at baseline	Years
PTEDUCAT	Years of education	Years
PTGENDER	Sex	Male/Female
PTMARRY	Marital status	Married/Widowed/etc.

Clinical Scales

Clinical Dementia Rating - Sum of Boxes (CDR-SB):

- Assesses 6 domains: memory, orientation, judgment, community affairs, home/hobbies, personal care
- Range: 0-18 (higher = worse)

- Sensitive to early changes

Functional Activities Questionnaire (FAQ):

- 10 items assessing instrumental ADLs
- Range: 0-30 (higher = more impairment)
- Completed by informant

Neuropsychological Tests

Test	Domain	Scoring
MMSE	Global cognition	0-30 (higher = better)
ADAS-Cog 11/13	Memory, language, praxis	0-70/85 (higher = worse)
RAVLT Immediate	Verbal learning	Sum of trials 1-5
RAVLT Learning	Learning rate	Trial 5 - Trial 1
RAVLT Forgetting	Retention	Trial 5 - Delayed
LDELTOTAL	Logical memory	Delayed recall score
TMTB	Executive function	Seconds (higher = worse)

ADAS-Cog in Detail

The Alzheimer's Disease Assessment Scale-Cognitive (ADAS-Cog):

```
# ADAS components in ADNIMERGE
adas_vars <- c("ADAS11", "ADAS13", "ADASQ4")

adnimerge |>
  filter(VISCODE == "bl", DX.bl == "LMCI") |>
  select(RID, all_of(adas_vars)) |>
  summary()
```

- **ADAS11**: Original 11-item version
- **ADAS13**: Adds delayed word recall and number cancellation
- **ADASQ4**: Word recall component only

RAVLT: Rey Auditory Verbal Learning Test

RAVLT Protocol:

- 15-word list presented 5 times (Trials 1-5)

- Interference list presented once
- Immediate recall after interference
- 30-minute delayed recall

Key derived scores:

- **RAVLT_immediate**: Sum of Trials 1-5 (learning capacity)
- **RAVLT_learning**: Trial 5 - Trial 1 (learning rate)
- **RAVLT_forgetting**: Trial 5 - Delayed (retention)
- **RAVLT_perc_forgetting**: % forgotten over delay

Feature Importance: Grassi et al.

Top predictors ranked by importance:

1. **ADAS13** - Most predictive overall
2. **CDR-SB** - Clinical staging
3. **RAVLT forgetting** - Memory retention
4. **FAQ** - Functional impairment
5. **LDELTOTAL** - Logical memory delayed recall
6. **TMTB** - Executive function
7. **MMSE** - Global cognition

Key insight: Memory retention (RAVLT forgetting) and executive function (TMTB) add predictive value beyond global measures like ADAS and MMSE.

Part V: Machine Learning Methods

Overview of ML Approach

Grassi et al. used an **ensemble** of 13 algorithms:

Ensemble Learning: Combining predictions from multiple models often outperforms any single model by reducing variance and capturing different aspects of the data.

The final prediction used **weighted rank averaging** across all models.

Category	Algorithms

Individual Algorithms Used

Category	Algorithms
Linear	Logistic Regression, Elastic Net
Probabilistic	Naive Bayes
Instance-based	k-Nearest Neighbors
Kernel	SVM (linear, polynomial, RBF, sigmoid)
Neural	Multilayer Perceptron
Tree-based	Random Forest, Gradient Tree Boosting

Random Forest: A Closer Look

```r
library(randomForest)

# Example: Train random forest for MCI conversion
rf_model <- randomForest(
  converter ~ ADAS13 + CDRSB + FAQ + RAVLT_forgetting +
              LDELTOTAL + TRABSCOR + MMSE,
  data = train_data,
  ntree = 500,
  mtry = 3,
  importance = TRUE
)

# Variable importance
importance(rf_model)
varImpPlot(rf_model)
```

Cross-Validation Strategy

Critical: Proper validation prevents overfitting and provides realistic performance estimates.

Grassi et al. used:

1. **Site-independent split**: Training and test sets from different ADNI sites
2. **5-fold cross-validation**: Within training set for hyperparameter tuning
3. **Held-out test set**: Final unbiased performance evaluation

Site-Independent Testing

```r
# Split by site to ensure independence
train_sites <- c("Site01", "Site02", "Site03", ...)
test_sites <- c("Site10", "Site11", "Site12", ...)

train_data <- adni_mci |> filter(SITE %in% train_sites)
test_data <- adni_mci |> filter(SITE %in% test_sites)
```

Site-independent testing simulates real-world generalization: Will the model work at a clinic that wasn't used for training?

Evaluation Metrics

Metric	Formula	Interpretation
Sensitivity	TP/(TP+FN)	Converters correctly identified
Specificity	TN/(TN+FP)	Stable correctly identified
PPV	TP/(TP+FP)	Precision for converters
NPV	TN/(TN+FN)	Precision for stable
AUROC	Area under ROC	Overall discrimination

ROC Curve Interpretation

```r
library(pROC)

# Calculate ROC curve
roc_obj <- roc(test_data$converter, predictions)

# Plot
plot(roc_obj, main = "MCI to AD Conversion Prediction")
auc(roc_obj)

# Optimal threshold
coords(roc_obj, "best", ret = c("threshold", "sensitivity", "specificity"))
```

Part VI: Project 2 Assignment

Project 2 Overview

Assignment: Develop a machine learning model to predict MCI to AD conversion using baseline ADNI data, following the Grassi et al. approach.

Due: March 3, 2026

Deliverables:

1. Statistical analysis plan (SAP)
2. R code in reproducible compendium
3. Written report (paper.Rmd)
4. Presentation (10 minutes)

Required Analysis Components

Your analysis must include:

1. **Data preparation**
    - Cohort selection (MCI subjects only)
    - Feature extraction (baseline only)
    - Missing data handling
2. **Model development**
    - At least 3 different algorithms
    - Cross-validation for tuning
    - Train/test split
3. **Evaluation**
    - ROC curves and AUROC
    - Sensitivity/specificity at chosen threshold
    - Feature importance analysis

Data Preparation Steps

```r
library(ADNIMERGE2)
library(tidyverse)

# Step 1: Load subject-level and cognitive data
data(ADSL)
data(ADQS)

# Step 2: Select MCI subjects at baseline
mci_baseline <- ADSL |>
  filter(DX_BL %in% c("EMCI", "LMCI"))

# Step 3: Determine conversion status
# (requires follow-up data - see next slide)
```

Defining Conversion Status

```r
# Get longitudinal diagnosis data
data(ADQS)

# Identify converters: any subsequent AD diagnosis
converters <- ADQS |>
  filter(USUBJID %in% mci_baseline$USUBJID,
         PARAMCD == "DX") |>
  group_by(USUBJID) |>
  summarize(
    converted = any(AVAL == "Dementia", na.rm = TRUE),
    time_to_conversion = min(ADY[AVAL == "Dementia"], na.rm = TRUE)
  )

# Merge back to baseline
mci_analysis <- mci_baseline |>
  left_join(converters, by = "USUBJID")
```

Suggested Timeline

Week	Task
Week 7 (Feb 17-21)	Data preparation, cohort definition
Week 8 (Feb 24-28)	Feature engineering, initial models

Week	Task
Week 9 (Mar 2-6)	Model tuning, evaluation, report
Mar 3	**Final submission**

Start early. Data cleaning and feature engineering take longer than expected.

Resources

Papers (on Canvas):

- Grassi et al. (2019) - Primary reference for ML approach
- Rye et al. (2022) - Additional ML approach with imaging
- Donohue et al. (2026) - ADNIMERGE2 and alzverse packages

R Packages:

- `ADNIMERGE2` - ADNI data (requires DUA from adni.loni.usc.edu)
- `caret` or `tidymodels` - ML framework
- `randomForest`, `glmnet`, `e1071` - Algorithms
- `pROC` - ROC analysis

Documentation:

- alzverse: atri-biostats.github.io/alzverse
- ADNIMERGE2: atri-biostats.github.io/ADNIMERGE2

Summary

Today we covered:

1. **ADNI**: Multicenter longitudinal study of AD biomarkers
2. **MCI**: Transitional state with ~36% 3-year conversion rate
3. **Grassi et al.**: Clinically-translatable ML approach (AUROC = 0.88)
4. **Key predictors**: ADAS13, CDR-SB, RAVLT, FAQ, LDELTOTAL, TMTB
5. **ML methods**: Ensemble of 13 algorithms with site-independent testing
6. **Project 2**: Replicate this approach in your research compendium

Next Steps

Before next class:

1. Apply for ADNI data access at adni.loni.usc.edu
2. Download ADNIMERGE2 package (requires Data Use Agreement)
3. Review Grassi et al. paper in detail
4. Begin cohort selection and data exploration
5. Draft your statistical analysis plan

References

Donohue MC, Hussen K, Langford O, Gallardo R, Jimenez-Maggiora G, Aisen PS. (2026). Alzheimer's clinical research data via R packages: The alzverse. *Alzheimer's Dement*, 22:e71152.

Grassi M, Perna G, Caldirola D, Schruers K, Duara R, Loewenstein DA. (2019). A Clinically-Translatable Machine Learning Algorithm for the Prediction of Alzheimer's Disease Conversion in Individuals with Mild and Premild Cognitive Impairment. *J Alzheimers Dis*, 67(4):1353-1365.

Petersen RC. (2004). Mild cognitive impairment as a diagnostic entity. *J Intern Med*, 256(3):183-194.

Rye I, Vik A, Kocinski M, Lundervold AS, Lundervold AJ. (2022). Predicting conversion to Alzheimer's disease in individuals with Mild Cognitive Impairment using clinically transferable features. *Sci Rep*, 12:15566.

Questions?

- **Office hours**: By appointment
- **Email**: rgthomas@ucsd.edu or maluo@ucsd.edu
- **Canvas**: Post in discussion board

Reminder: Project 2 presentations begin March 3. Prepare a 10-minute summary of your analysis approach and findings.